

Notes 2

1. What is an Operating System?

An operating system provides all fundamental software features of a computer. An OS enables you to use the computer's hardware providing you the basic tools that make the computer useful. All of those features rely on the OS's kernel. Other OS features are owed to additional programs that run atop the kernel.

2. What is a kernel?

An OS kernel is a software component that's responsible for managing low-level features of the computer, including the following managing system hardware, memory allocation, CPU time, and program to program interaction.

3. Which other parts aside from the kernel identify an OS?

The kernel is at the core of any OS, but it's a component that most users don't directly manipulate. Instead, most users interact with a number of other software components. Such programs include the following:

- Command-Line Shells

This was the de facto way of using computers before the Graphical Interface was invented. CMDs work by typing commands in a shell. In Linux, the entire system can be control via the CLI.

- Graphical User Interfaces

GUIs rely on icons, menus, and a mouse pointer for the user interaction. Linux relies on a GUI known as the X Window System in combination with desktop environment program suites.

- Utility and Productivity Programs

Tools like web browsers, document processors and text editors.

- Libraries

Libraries are collections of programming functions that can be used by a variety of programs.

4. What is Linux?

- Linux is a Unix-like Operating System popular in academic and business environments.
- Linux consists of a kernel, libraries, and utilities that make up the entire operating system.
- Linux is available in many distributions (any operating system that runs the Linux kernel)
- Popular Linux distributions include Arch, CentOS, Debian, Fedora, openSUSE, Red Hat, Slackware, Ubuntu and many more.
- Linux has several characteristics that make it worth learning and using:

\$\circ\$ Linux is open source software

\$\circ\$ Linux is available free of charge.

\$\circ\$ Linux includes many of the Unix tools including many important Internet server programs and programming languages out of the box

\$\circ\$ Linux is highly scalable and customizable

\$\circ\$ Many businesses and nonprofit organizations rely on Linux for their day to day operations

\$\circ\$ The vast majority of server applications on the internet run on some version of Linux

\$\circ\$ You can install Linux on almost any system as it supports almost every processor architecture

5. What is a Linux distribution?

- Debian

\$\circ\$ Ubuntu

\$\circ\$ Pop OS

- Redhat

\$\circ\$ Fedora

\$\circ\$ Alma Linux

There are other independent distros like:

- Slackware

\$\circ\$ Absolute Linux

- Arch Linux

\$\circ\$ Manjaro

\$\circ\$ SteamOS

6. List at least 4 Linux characteristics:

- Linux is open source
- Its highly secure
- Its stable for long term running
- Linux could run on a lot of hardware

7. What is Debian?

- Debian is an all-volunteer organization dedicated to developing free software and promoting the ideals of the Free Software community.
- The Debian Project began in 1993, when Ian Murdock compiled a group of developers to create a coherent linux distribution

- Debian is known to be the Grandfather of all linux distributions alongside Slackware and Opensuse.
- Ubuntu was created to be a user friendly version of Debian and with time it has become a Major distribution of its own with other Distribution using it as its base.

8. List and define the different types of licensing agreements

- Software is a type of intellectual property that is governed by copyright laws and, in some countries, patent laws.
- Open source software, however, relies on licenses, which are documents that alter the terms under which the software is released.
- Types of licensing agreement:

Open Source: the software may be distributed for a fee or free. The source code is distributed with the software.

Closed Source: the software is not distributed with the source code. The user is restricted from modifying the code.

- Freeware: the software is free but the source code is not available.
- Shareware: the software is free on a trial basis.

Free software the software is distributed with the source code. The software can be free of charge or obtained by a fee.

9. What is Free Software? Define the 4 freedoms.

Freedom 0: use the software for any purpose

Freedom 1: examine the source code and modify it as you see fit

Freedom 2: redistribute the software

Freedom 3: redistribute your modified software

10. What is virtualization?

- Virtualization is defined as creating virtual versions of something.
- Virtualization is often used to let multiple OSs run on one physical machine at the same time.
- Virtualization allows administrators to divide the hardware and create multiple computers inside a single physical computer.
- Virtualization is an old concept however it has gained popularity due to the availability of faster, better and cheaper hardware.
- Virtualization is one of the corner stop technologies of Cloud computing.